

The Last Mile of Data

The work happens outside.
The systems are inside.
Nothing connects them.

A contractor runs crews across dozens of sites. The office has good systems. **The field runs on paper.**

START HERE

- 1. **The agents** — five to build, four things that must not be.
- 2. **The build** — three hours, hour by hour.
- 3. **Setup** — do this before the clock starts.

MORE DETAIL

Why anyone should care

01

Nobody can be scored

You cannot score a crew you have no records for. Every quality measure ships blank.

02

Nothing can be planned

You cannot forecast without knowing how fast crews actually work. Every date is a guess.

03

Disputes are lost by default

When the customer queries a number months later, the answer is a spreadsheet cell. That is not evidence.

One missing fact. Three dead screens. **Fix capture and all three light up.**

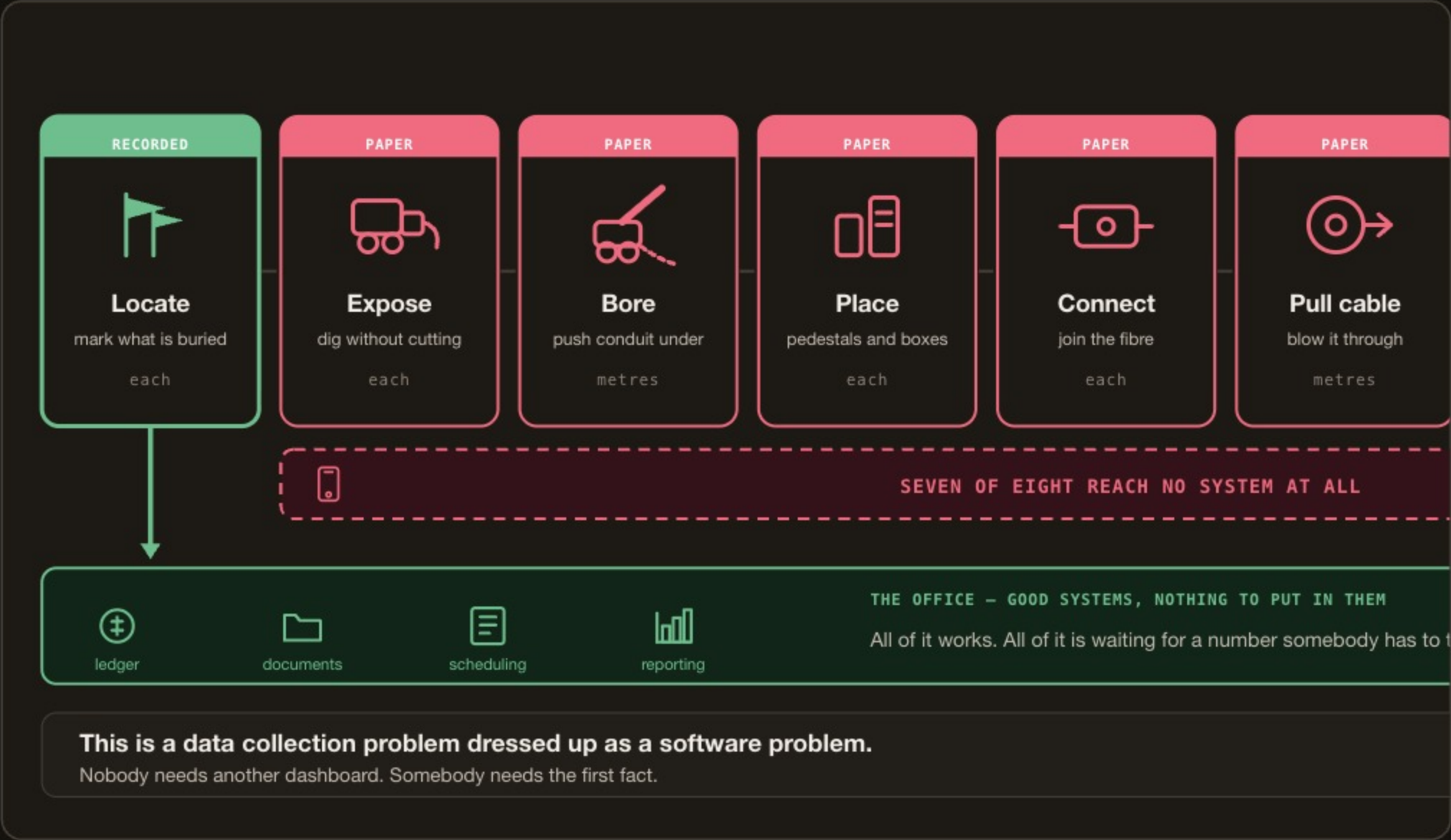
The value hypothesis, stated so it can be measured

IF CAPTURE WORKS	HOW YOU WOULD KNOW WITHIN ONE SEASON
Billing accuracy improves	Run capture in parallel with paper for one crew. Measure the disagreement rate between the two, and which one turns out to be right.
The office finds out sooner	Median days from work performed to work recorded. Today that is somewhere between one and twenty-one. Target: same shift.
Disputes become answerable	Proportion of billed line items that can produce a timestamped, located, attributed record on request. Today: none.
Rework becomes visible	Rework is currently invisible because nothing records the first attempt. Count of rework events per thousand units, which is a number nobody can currently produce.

THE DISCIPLINE

Do not promise productivity. Measured studies repeatedly find people believe tooling made them faster when it did not. **Promise the fact, and measure the fact.**

Eight stages. One gets recorded.



MORE DETAIL

The pattern is not specific to one trade. Substitute pest control, solar installation, road maintenance, tree work, HVAC service, agricultural contracting or disaster restoration and the shape holds: **dispersed crews, physical work, a paper record, and an office that finds out weeks later.**

WHAT ACTUALLY HAPPENS TODAY

A crew finishes work. The foreman writes quantities on a paper ticket. The ticket travels in a truck. Somewhere between one day and three weeks later, somebody in an office types it into a spreadsheet. By then nobody can check it.

WHY IT HAS NOT BEEN SOLVED ALREADY

Because every attempt starts at the office. Dashboards, reports, integrations, a data warehouse. All of them assume the fact exists. **The fact does not exist.**

Capture it. Organise it. Use it.

ONE • CAPTURE

Get the fact

Someone outside, in gloves, after ten hours, with no signal. Something has to leave as data.

Could a foreman file a day’s work in thirty seconds without typing a sentence?

TWO • ORGANISE

Make it mean something

A photo with no crew, site or work type is just a photo. Joining it to the plan is the job.

Can you reconstruct months later who decided, and on what evidence?

THREE • USE

Change a decision

Reports they could never run before: what got built, by whom, when, against what was sold.

Does it change what somebody does on Monday?

A small complete loop beats a big broken one. **One work type, one crew, all three steps.**

MORE DETAIL

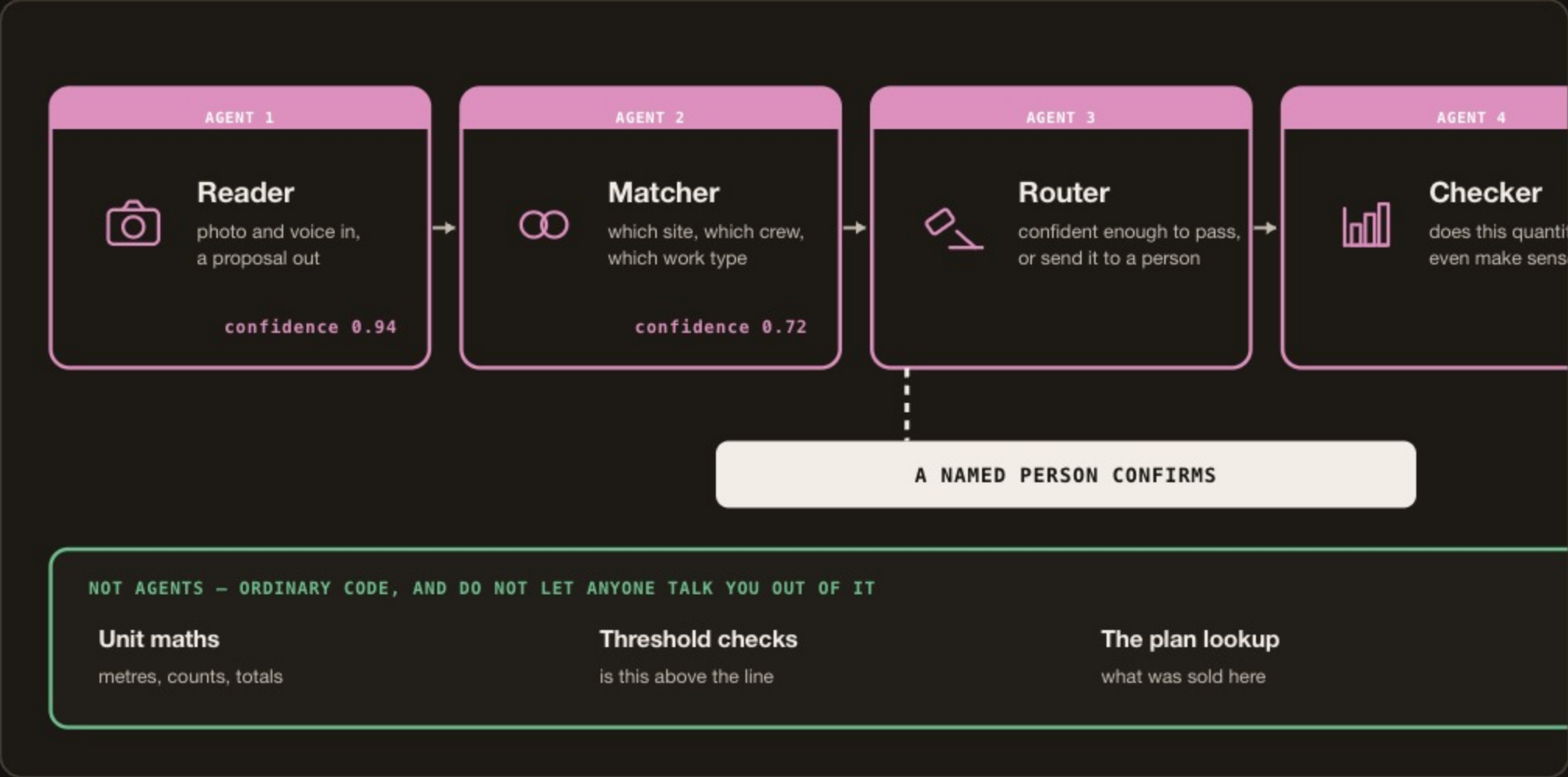
WHY ALL THREE, AND NOT JUST THE FUN ONE

Capture alone is a form. Organise alone is a schema nobody filled in. Use alone is a dashboard of invented numbers. **The demo that lands is the one where a thing that happened outdoors becomes a number somebody would act on,** end to end, however narrow.

THE QUESTION TO SETTLE IN YOUR FIRST TWENTY MINUTES

Is the unit of capture the *crew-day* or the *piece of work*? A crew-day is easy to file and hard to bill against. A piece of work is precise and tedious to record. Most real systems land on a crew-day carrying a list of pieces — but decide out loud and early, because everything downstream inherits it.

Five agents. And four things that must not be.



MORE DETAIL

THE RULE THAT WILL SAVE YOU THREE HOURS

A model decides **what something is**. Code decides **whether it is allowed, whether the numbers add up, and what gets written**.

Every team puts a model somewhere it does not belong, usually in the arithmetic. If a competent developer could write the rule down in an afternoon without listing examples, it is not an agent. It is an `if`.

Each agent, and what makes it hard

AGENT	IN	THE HARD PART
Reader	A photo of a counter, or thirty seconds of speech	Getting a number off a dirty LED display at an angle. Easy at 80%, painful at 98%. Decide early which you need.
Matcher	Position, time, whatever the person said	This is the real project. "The north run past the bridge" has to become a site id. Get it wrong and every downstream number is attached to the wrong job.
Router	A proposal and a confidence	Not the model — the thresholds. Route by what it costs to be wrong, not by confidence alone.
Checker	The proposed quantity and the plan	Mostly arithmetic. A crew cannot bore 4,000 metres in a day. Catching that needs no model at all.
Answerer	A question and the confirmed facts	Tempting and low value. Build it last, or not at all. A chart beats a chatbot for an owner at 6am.

CHAIN LENGTH

Five agents in a row at 90% each is 59% end to end. **Put a deterministic check between every pair**, and count the arrows in your design that have no check on them. That number is your real risk.

Three hours. Four people. One loop.

CLOCK	WHOEVER TAKES CAPTURE	WHOEVER TAKES THE SPINE	WHOEVER TAKES THE DEMO
0:00–0:30	Phone page that takes a photo and stores it locally	Load the five CSVs. One table for proposals.	Pick the one report. Write the sentence you will say.
0:30–1:15	Reader: photo in, number out	Matcher: position and time to a site id	Stub the report off the CSVs so it exists early
1:15–2:00	Offline queue. Kill the wifi and prove it.	Router plus the confirm screen	Wire the report to confirmed facts only
2:00–2:30	The return screen. Yesterday, and what it is worth.	Checker. Refuse one thing on purpose.	Rehearse. Out loud. Twice.
2:30–3:00	Everyone stops building. Run the whole loop end to end, on a phone, in the room, from photo to report.		

Six hours? Same plan. **Use the extra three to make it more real, not prettier.** Real footage, a second crew, a measured comparison against the paper.

MORE DETAIL

THE DEMO YOU ARE AIMING AT, IN ONE PARAGRAPH

Someone walks to the back of the room, photographs a counter, and taps once. Thirty seconds later a name appears on a queue, somebody confirms it, and a bar on a chart moves. **That is the whole pitch.** If you can do that with one work type, you have out-demoed anyone with six screens and no loop.

IF YOU ARE TWO PEOPLE, NOT FOUR

Cut the Checker and the Answerer. Cut the return screen to a text message. Keep capture, matcher, confirm and one chart. The loop is what matters; the breadth is not.

THE HOUR YOU WILL BE TEMPTED TO WASTE

Hour two, on the matcher, trying to make it clever. **Make it dumb and correct instead:** nearest site from a list of twelve, with a confirm step. Cleverness there wins nothing and costs everything.

Do this before the clock starts.

- **A phone that can reach your laptop.**
Same wifi, or a tunnel. Fifteen minutes of faff that will cost you an hour if you leave it to the day.
- **The five CSVs loaded, in something you can query.**
Copy them off the data slide now. Do not spend hackathon minutes on a schema.
- **Model access working, with a photo already sent through it.**
One round trip, proven, before you are under pressure.
- **Something to photograph.**
A number on any display — a kitchen scale, a meter, a printed digit on a card. You need a test subject that is not a screenshot.
- **One page where a fact can appear.**
Even a list. It is where everything ends up and it is oddly easy to forget to build.

MORE DETAIL

STACK ADVICE, HELD LIGHTLY

Use what you already know. A hackathon is not the place to learn a framework, and the judges cannot see what it is written in. The only two choices that actually matter: **where the photo goes**, and **whether the phone works with the network off**. Settle both in the first thirty minutes.

THE FIVE-MINUTE TEST THAT PREDICTS YOUR DAY

Before anything else: take a photo on the phone, get it onto the laptop, and print its size to a console. If that takes more than fifteen minutes, your architecture is wrong and it is cheap to find out now.

WHAT TO HAVE READY TO SAY

One sentence on the problem, one on what you built, one on what it would take to run it for real. **Write them before you start building**, because you will not have the clarity afterwards and every team discovers this at the same moment.

Go film it. Here is the shot list.

You cannot design this from a desk. Get footage of each stage — filmed, downloaded or staged.

THE BEST IDEA IN THIS BRIEF, AND IT COMES FROM THE TRADE

At four of the eight stages, **the number is already on a screen**. Drill rigs count rods. Cable machines count metres. Test sets show readings.

So maybe nobody reports anything. Maybe you just need **a photo of a counter** and something that reads it.

MORE DETAIL

Stage by stage: what to shoot, and what it would give you

STAGE	THE SHOT	WHAT A MACHINE COULD TAKE FROM IT
Locate & mark	Wide shot down a street of coloured paint marks and flags; a close-up of one flag	Which utilities are present, from the colours. Area covered. The date. Whether the marks are fresh or weathered.
Expose	The vacuum truck, the hose in the hole, the exposed utility at the bottom	Hole count. Depth against the tape. What was found and whether it matched the plan.
Bore	The rig’s rod rack and the operator’s display; entry and exit pits	Metres, from rod count × rod length. This is the disputed number, and it is sitting on a screen.
Place	The pedestal or box going into the ground, with the house number in frame	Count, and the civic address it serves — which is the join to the plan.
Connect	The open enclosure, the fibres inside, the label on the door	Which box. How many connections. Whether it was left tidy, which is a quality signal nobody records.
Pull cable	The blowing machine’s metre counter; the spool before and after	Metres, read straight off the counter. Two photos bracket the run.
Verify	The test set screen showing the trace and loss figures	Pass or fail, and the actual readings. This is the acceptance evidence the customer will later ask for.
Restore	The same frame before work and after — sod, asphalt, curb	Area restored. A before-and-after pair is the entire dispute defence for surface damage.

THE ONE SHOT YOU MUST NOT SKIP

Film the paper ticket being filled in — a foreman at the tailgate, at the end of the day, writing numbers on a page. That is the incumbent. It is fast, it never crashes, and everybody already knows it. Your demo is judged against it, so put it on screen and let the room see what you are competing with.

IF YOU CANNOT GET ON A SITE

Footage of directional drilling, vacuum excavation, cable blowing and fibre splicing is abundant and public — manufacturers, trade channels and operators all post it. Pull stills from that. For the paper ticket and the tailgate moment, stage it. Ten minutes with a clipboard in a car park is enough and it will be the most useful asset you have.

WHAT TO BRING BACK

Per stage: two or three stills and one short clip. Plus one continuous walk down a street, phone held at chest height, narrating what you see. That last one is your test input for everything — voice, vision, geolocation and the awkward reality of a moving camera, all in one file.

Fake data. Real shape. Copy it.

Five files. Same numbers for every team. All invented, so share them freely.

daily_production.csv is the one that matters. In the real world it does not exist. That is what you are building. Treat it as the answer key, not the input.

MORE DETAIL

What is in each file

FILE	ROWS	WHAT IT IS
sites.csv	12	Where the work happens. A site code, a town, how many homes, when it started, and its status.
crews.csv	8	Eight crews across three subcontractors. A foreman, a headcount, and the work types each crew is certified for. Note that crews belong to subcontractors, not to the contractor.
work_types.csv	8	The eight stages, their units, and a revenue rate. Units matter: some are counts and some are metres.
planned.csv	96	Planned quantity per site per work type. This is what was sold. The denominator.
daily_production.csv	185	Eight weeks of crew-days. Date, crew, site, work type, quantity, how it was captured, and <code>days_to_office</code> — the lag before anybody in a building knew.

THE LAG COLUMN IS THE WHOLE BRIEF IN ONE FIELD

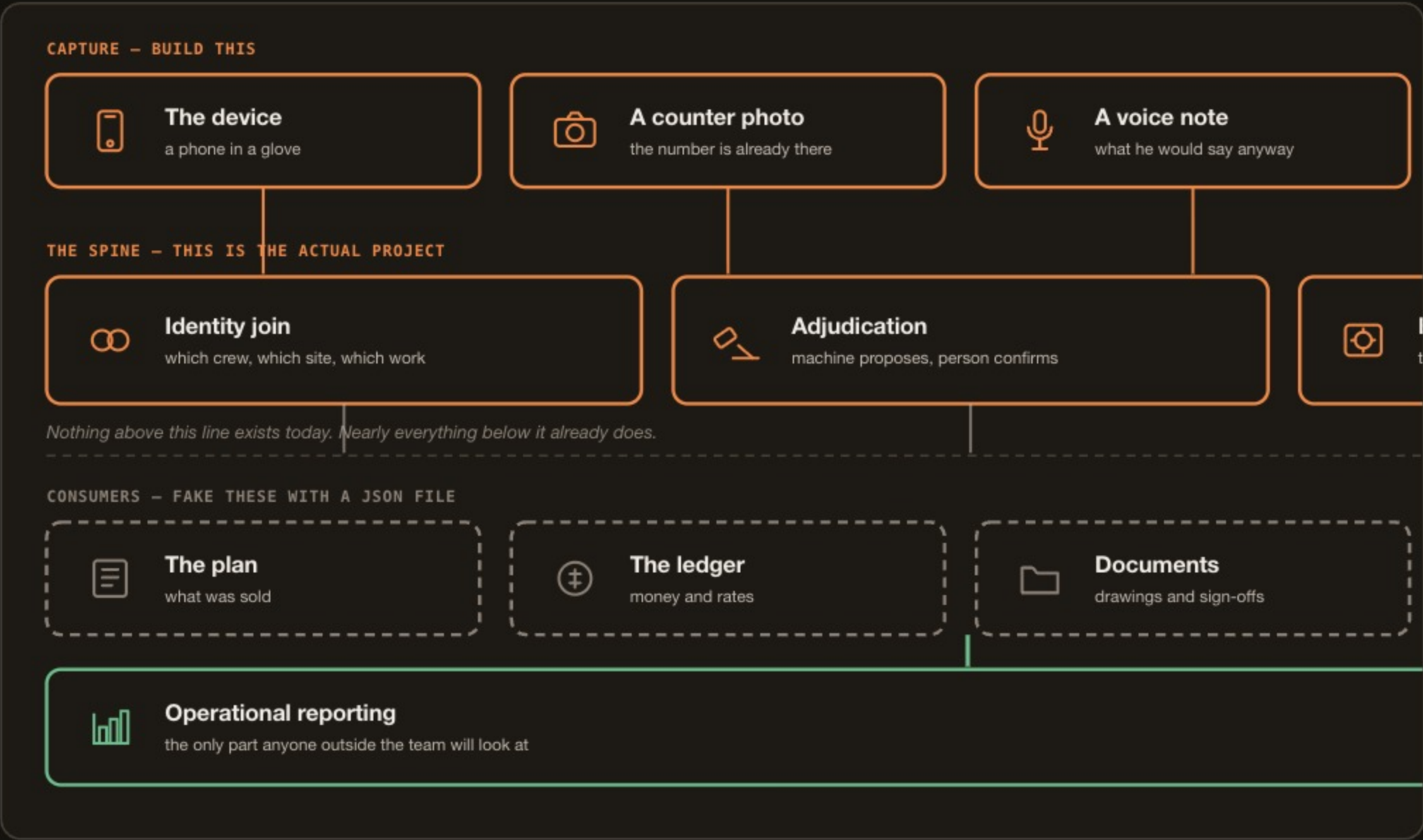
Across 185 crew-days: 25 reached the office next day, 52 took a week, 32 took a fortnight, and 29 took three weeks. **Plot that distribution first.** It is the most persuasive chart in the dataset and it takes four lines of code.

WHAT THE DATA DELIBERATELY DOES NOT GIVE YOU

No geolocation. No photographs. No timestamps finer than a date. No crew member names. No evidence of any kind.

That is not an oversight. **Those are the things your solution has to invent a way to collect** — and deciding which of them are worth the friction is most of the design work.

Build three things. Fake the rest.



MORE DETAIL

THE SCOPING RULE

Anything below the dashed line already exists at a real company, in some form, and rebuilding it in a hackathon proves nothing. **Stub it with a JSON file and spend your hours on the spine.** The identity join is the hardest and least glamorous of the three, and it is the one that decides whether any of it works.

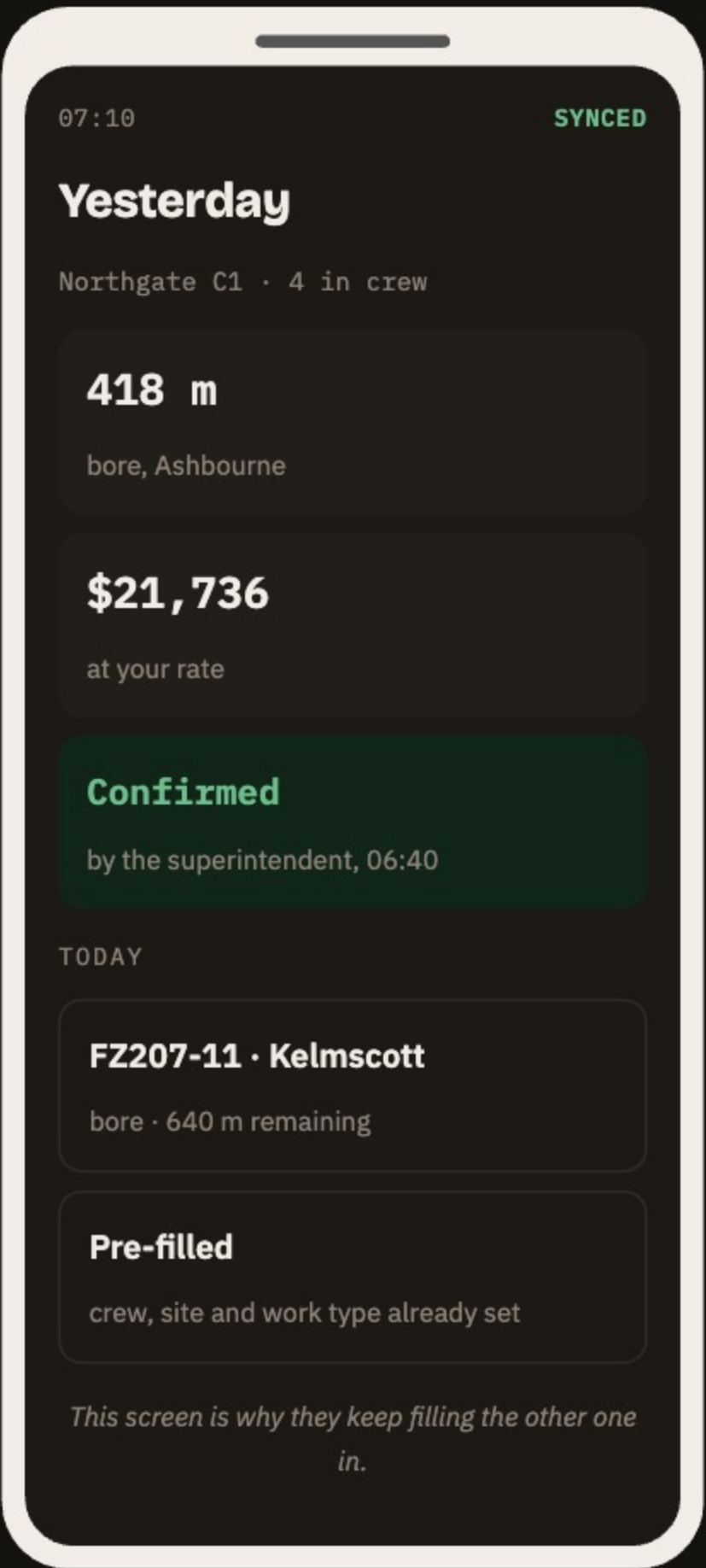
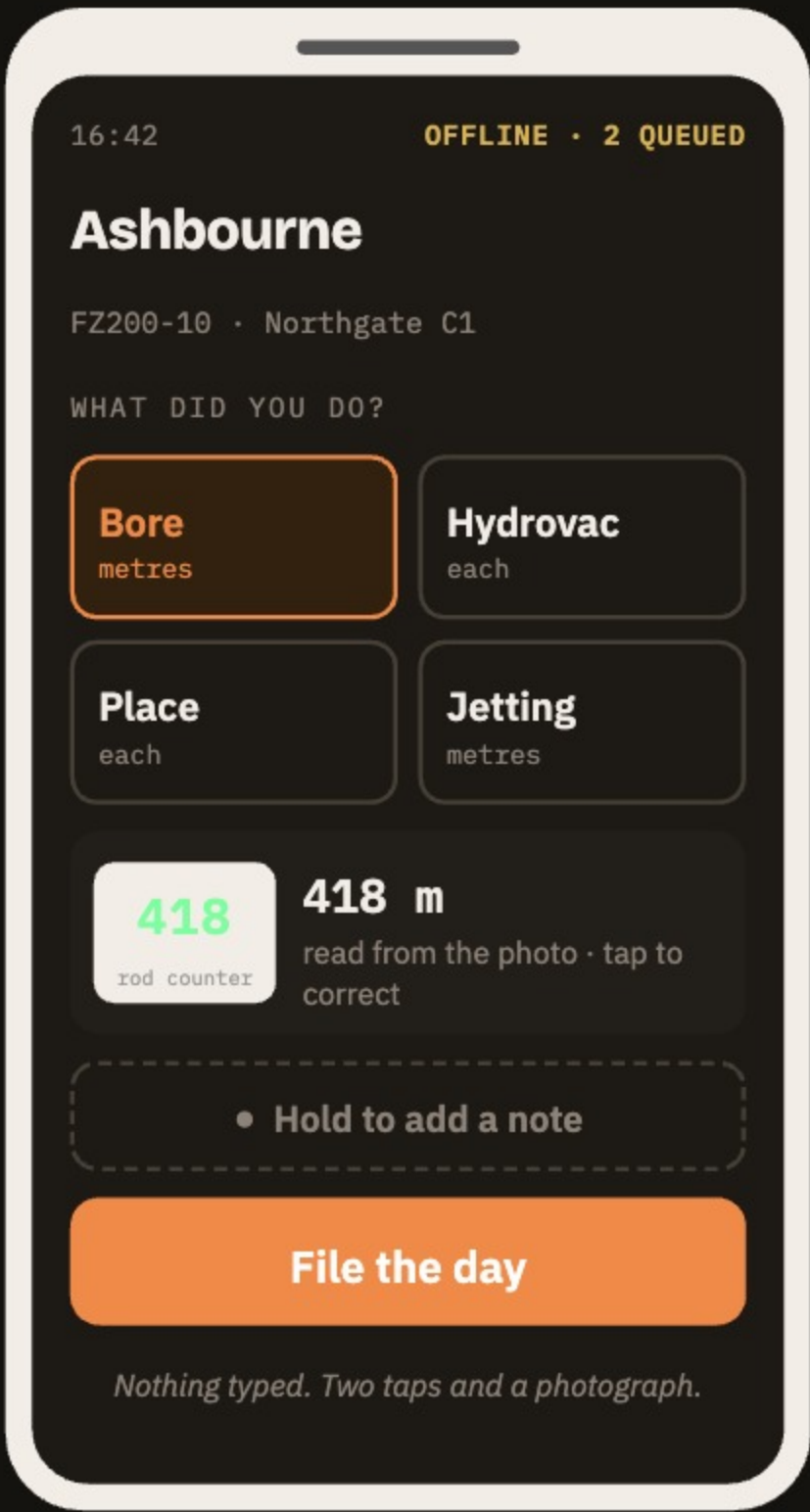
What each integration would really involve

CONNECTION	IN A HACKATHON	IN REALITY
Device → spine	Build it	Offline queue, retry, conflicting submissions, a device that may belong to someone who does not work for you.
Media → evidence store	Build it	Object storage, content hashing, and a decision about whether a photo is evidence or just an attachment. They are different things with different retention.
Spine → the plan	Stub: <code>planned.csv</code>	The plan lives in whatever the contract was written in. Usually a spreadsheet. Sometimes a PDF.
Spine → the ledger	Stub: a printed line	An accounting API, usually read-only until somebody signs off on letting a machine write. That sign-off takes longer than the integration.
Spine → documents	Stub: a folder	A document store with a folder convention older than the software, which nobody will let you change.
Spine → customer	Stub: a PDF	The customer dictates the billing format and usually has no integration at all. Expect email.

THE THING THAT SURPRISES PEOPLE

In field-services businesses the integration difficulty is rarely technical. **The counts are small and the protocols are boring.** The difficulty is that most of the connections are a person reading something and typing it somewhere — and each of those people has a routine that works and a reason not to change it.

Two screens nobody has built yet.



Why these two go together

Field software is abandoned because the person filling it in gets nothing back. **Build the right-hand screen first.** Then the left-hand one has a reason to exist.

Note what is absent from both: a keyboard, a form, a dropdown of forty sites, and any word longer than one syllable.

The counter photo

The quantity is not typed. It is read off a machine that already counted it, which is faster, harder to fudge, and carries its own evidence.

MORE DETAIL

What exists today, and what these replace

TODAY	THE MOCKUP
A paper ticket at the tailgate	Two taps and a photograph, filed before the truck leaves. Offline by default; the queue drains whenever signal returns.
A spreadsheet typed up days later	The fact is created at the place and time it happened, with the device’s clock and position attached.
Nothing goes back to the crew	Yesterday’s metres, what they are worth, and who confirmed them — before the crew leaves the yard this morning.
No crew identity anywhere	The crew is the session. Every fact is attributed without anyone being asked who they are.

THE DESIGN ARGUMENT WORTH HAVING

The left screen assumes the crew captures. An alternative worth prototyping: **the crew captures nothing**, and a supervisor walking the line at the end of the day photographs four counters and files for everyone. Fewer people to convince, one device to manage, and the same facts. It is less elegant and it might be right.

A machine proposes. A person signs.

Adjudication queue

7 waiting · 2 refused

Northgate C1

bore · 418 m

0.94

Halloway C2

hydrovac · 6

0.91

Prentiss C1

jetting · 512 m

0.72

Halloway C1

site unresolved

refuse

418

counter, 16:41

CREW

Northgate C1 · Foreman A1

SITE

FZ200-10 Ashbourne

matched on position + plan

WORK

Directional bore

QUANTITY

418 m

read from counter · 0.94

CHECK

Within plan

4,118 planned, 3,240 done

Confirm

Refuse...

Refusing asks for a reason code. Refusals are kept, not deleted.

Ashbourne FZ200-10 — actual against sold

updated 4 minutes ago

Locates

62 / 74

Hydrovac

19 / 27

Directional bore

3,240 / 4,118

Pedestals

148 / 206

Ground boxes

9 / 25

Cable jetting

1,980 / 3,724

Splicing

44 / 186

Restoration

310 / 882

Every bar is the sum of confirmed crew-days. Click one and you get the days. Click a day and you get the photograph.

MORE DETAIL

The third screen: one number, fully traced

Evidence pack — invoice line 4471

exported 19 Sep

The claim

418 m directional bore, 14 July

The capture

Photo, 16:41, device clock,
position within 8 m of the site

The proposal

Machine read 418 from the
counter, confidence 0.94

The verdict

Confirmed by the superintendent, 06:40 the next morning

The fact

Billable, and defensible eight months later

WHAT MAKES THIS DIFFERENT FROM A FORM AND A REPORT

Every number carries the thing that produced it. **The refusals are kept beside the confirmations**, with their reasons, so the record shows judgement being exercised rather than data being entered. That is what turns a quantity into evidence.

THE FAILURE MODE TO DESIGN AGAINST

Approval fatigue. A queue of hundreds at 0.9-plus confidence trains the reviewer to tap Confirm without looking, and then the signature means nothing. **Route by cost of error, not by confidence alone**, audit a random sample of the auto-approved, and treat a rising approval rate as a defect in the rules rather than diligence in the reviewer.

What you can report once you have the facts.

DAILY

Who did what, where

By end of shift, not end of month. Who worked where, on what, and which sites nobody visited.

WEEKLY

Actual against sold

Done versus sold, per site, per work type. Their first honest burn-down.

ON DEMAND

Defend a number

Any invoice line traced back to a day, a crew and a photo. The one the customer asks for later.

Then the ones nobody can run today: **crew productivity, rework rate, and a finish date you can calculate.**

MORE DETAIL

Six reports, and what each one needs to exist

REPORT	NEEDS	WHY IT CANNOT BE PRODUCED TODAY
Daily activity	crew × date × site	The record arrives days or weeks later, so "today" is never knowable today.
Production against plan	quantity × work type × site	Quantities exist, but typed in arrears with no way to check them, so nobody trusts the variance.
Crew productivity	quantity per crew-day	There is no crew identifier anywhere. Output cannot be attributed to the people who produced it.
Rework rate	a second visit linked to a first	Nothing records the first attempt, so a repeat looks identical to new work.
Evidence pack	a fact plus its provenance	The record is a spreadsheet cell. It has no source, no timestamp and no author.
Forecast finish	observed rate × remaining	No observed rate exists, so every forecast is the original plan restated with more confidence.

BUILD ONE OF THESE, NOT SIX

Pick the report that would change a decision on Monday and work backwards from it to the capture. **Crew productivity is the most tempting and the most dangerous** — the moment a number can rank people, the people being ranked start managing the number. Production against plan is the safer first report and the one an owner will actually open.

A FAIR WARNING ABOUT THE DEMO

Reports are the easiest thing to fake and the judges know it. A chart built from the synthetic file proves nothing. **A chart built from a fact your own capture created ten minutes ago, in the room, proves everything.** Even if it is one row.

Five things that kill field apps.

- **The people doing the work do not work for you.**
Subcontractors. No company email, no login, no reason to learn your app.
- **Nothing goes back to them.**
The crew fills in the form and gets nothing. This is why field apps get abandoned.
- **There is nothing to attach a fact to.**
No crew ids, no site ids. Most of the work is joining, not capturing.
- **Conditions are bad.**
Rain, gloves, glare, a dead battery, no signal, and someone who wants to go home.
- **Paper works.**
It is fast, it never crashes, everyone knows it. You are competing with a pen.

MORE DETAIL

THE TRAP NEARLY EVERY TEAM FALLS INTO

Building the office end. It is the comfortable half — dashboards, charts, a clean data model, a satisfying demo. **It is also the half that already exists at most companies and is not the problem.** If by the halfway point your demo only shows screens somebody in an office would look at, you have built the wrong thing.

THE QUESTION WORTH ARGUING ABOUT IN YOUR TEAM

Is the unit of capture the *crew-day* or the *piece of work*? A crew-day is easy to file and hard to bill against. A piece of work is precise and tedious to record. Most real systems end up with a crew-day carrying a list of pieces — but decide deliberately, early, and out loud, because everything downstream inherits it.

And four things not to build

A coordination product

Scheduling, dispatch, managing subs. Good products exist. Nobody asked for another.

An accounting integration

The accounting works fine. Assume the number gets there somehow. The problem is upstream.

A general reporting layer

Reports come after facts. If you are designing filters, you have drifted.

Anything needing a rollout

If it needs training and managed devices, it will never get tried. Assume one willing foreman.

WHY THE ANTI-GOALS ARE HALF THE BRIEF

Every one of these is more comfortable to build than the actual problem, and each has consumed a real programme somewhere. The discipline of a short build is choosing what not to touch, and the teams that do best usually say out loud, early, what they are refusing to do.

How far you got. Not how it looks.

The winner is whoever travels furthest — from a vague problem to something demonstrated.

ARTIFACT	WHAT IT HAS TO DO
The problem	Name who experiences it and what happens to them today. Specific enough that somebody could disagree with you.
The why	What becomes possible, and how it would be measured. A number you could check in one season.
The data	What sources you would need and what has to be true to trust them. Including the ones that do not exist yet.
The proof	A working thing. It can be narrow. It cannot be a mockup of a thing.
The next step	One page: owner, the experiment, the ask, what it would prove.

MORE DETAIL

THINGS THAT EARN CREDIT

Handling a case where the system should refuse. Showing a number you measured rather than asserted. Demonstrating it in conditions that resemble the real ones. Admitting what you did not build and why. Using the synthetic data honestly, including the awkward rows.

THINGS THAT DO NOT

A prettier interface. More screens. An integration to a product nobody named. A chart of invented numbers. Claiming a percentage improvement you did not measure. Machine learning where a rule would do.

IF YOU ONLY HAVE THREE HOURS

Build the capture, on one work type, for one crew, offline, with a return path. That is a complete loop at the smallest possible scale, and a complete small loop beats an incomplete big one every time.

ABOUT THIS BRIEF

What this is.

A problem brief for a Milbird hackathon.

No client data is in here. Every number invented, every name made up. The trade terms are standard across the industry.

Share freely.

MORE DETAIL

The shape of the problem — competent back office, absent field capture, a paper record travelling in a truck — is common enough across field-services industries to be worth treating as a pattern rather than a case. That is why this brief names no company, no product and no place.

The eight stages use industry-standard terms that thousands of contractors use daily. The towns, crews, subcontractors, site codes, rates and quantities are all invented. The lag distribution is plausible rather than observed.

If your solution turns out to be good, the interesting question is which other trade it transfers to first.